

HANDS-ON 10 – Unit Testing Angular Applications (Advanced)

Aim

To learn how to test Angular components, services, and NgRx store using Jasmine, Karma, TestBed, HttpClientTestingModule, and MockStore.

Objective

- Test Angular components using Jasmine and TestBed.
- Test @Input and @Output properties.
- Test HTTP services using HttpClientTestingModule.
- Test NgRx store-connected components using MockStore.

Task 1: Testing CourseCardComponent

Step 101

- Configure TestBed for CourseCardComponent.
- Create component and fixture.

Step 102

- Write a test to verify the component is created successfully.

Step 103

- Test the @Input() property.
- Verify that the course name is displayed correctly.

Step 104

- Test the @Output() event.
- Verify the enroll event is emitted when the button is clicked.

Step 105

- Test ngOnChanges().
- Verify the expected method or console log is executed.

Task 2: Testing CourseService and NgRx Store

Step 106

- Configure HttpClientTestingModule.
- Inject CourseService and HttpTestingController.

Step 107

- Test getCourses() API.
- Verify the correct API URL is called.

- Check the returned course data.

Step 108

- Test HTTP error handling.
- Verify error response is handled correctly.

Step 109

- Configure MockStore.
- Test a component connected to the NgRx Store.
- Verify course data is displayed from the store.

Step 110

- Change the store state using setState().
- Verify the loading indicator is displayed.

Outputs:

The screenshot shows the VS Code interface with the Angular HandsOn project. The Explorer panel on the left shows the project structure, including the 'student-course-portal' folder and its subdirectories. The main editor displays the 'course-list.ts' file, which imports various Angular and RxJS modules and defines the 'CourseList' component. The terminal panel at the bottom shows the output of running 'npm test', which includes the following text:

```
PS C:\Users\HP\Desktop\Deepskill\Angular_HandsOn\student-course-portal> npm test -- --watch=false
✓ student-course-portal src/app/components/header/header.spec.ts (1 test) 52ms
✓ student-course-portal src/app/services/enrollment.spec.ts (1 test) 7ms
✓ student-course-portal src/app/directives/highlight.spec.ts (1 test) 2ms
✓ student-course-portal src/app/pipes/credit-label-pipe.spec.ts (1 test) 2ms
stdout | src/app/pages/course-list/course-list.spec.ts > CourseList > should render course cards from MockStore state
Course changed: Data Structures
Course changed: Angular Advanced
stdout | src/app/pages/course-list/course-list.spec.ts > CourseList > should show loading indicator when store loading is true
Course changed: Data Structures
Course changed: Angular Advanced
✓ student-course-portal src/app/pages/course-list/course-list.spec.ts (3 tests) 882ms
✓ should create 665ms

Test Files 14 passed (14)
Tests 23 passed (23)
Start at 09:19:01
Duration 9.58s (transform 1.68s, setup 7.10s, import 3.14s, tests 2.57s, environment 76.11s)

PASS Waiting for file changes...
press h to show help, press q to quit
```

FileEditSelectionViewGoRunTerminalHelpAngular_HandsOn2

EXPLORER

ANGULAR_HANDSON

student-course-portal

src

app

components

notification

directives

features

guards

interceptors

models

course.model.ts

enrollment.model.ts

student.model.ts

pages

course-detail

course-list

course-list.css

course-list.html

course-list.spect.ts

course-list.ts

courses-layout

enrollment-form

home

home.css

home.html

home.spect.ts

home.ts

not-found

reactive-enrollment-form

student-profile

student-profile.css

TIMELINE

master

1

0

TS course-list.ts U

course-list.html U

course-list.css U

home.ts U

TS student

student-course-portal > src > app > pages > course-list > TS course-list.ts > ...

1 import { Component, OnInit } from '@angular/core';

2 import { CommonModule } from '@angular/common';

3 import { FormsModule } from '@angular/forms';

4 import { ActivatedRoute, Router } from '@angular/router';

5 import { Store } from '@ngrx/store';

6 import { BehaviorSubject, Observable, combineLatest, map } from 'rxjs';

7

8 import { CourseCard } from '../../components/course-card/course-card';

9 import { CourseService } from '../../services/course';

10 import { Course } from '../../models/course.model';

11 import { loadCourses } from '../../store/course/course.actions';

12 import {

13 selectAllCourses,

14 selectCoursesError,

15 selectCoursesLoading

16 } from '../../store/course/course.selectors';

17 import { loadEnrolledCourses } from '../../store/enrollment/enrollment.actions';

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

powershell - student-course-portal

PS C:\Users\HP\Desktop\Deepskilling\Angular_HandsOn\student-course-portal> npm run build

> ng build

Initial chunk files | Names | Raw size | Estimated transfer size

chunk-DFNP1QFM.js | - | 224.94 kB | 58.97 kB

main-YM2H2HNW.js | main | 137.30 kB | 34.87 kB

styles-5I1URTS0.css | styles | 0 bytes | 0 bytes

Initial total | 362.24 kB | 93.84 kB

Lazy chunk files | Names | Raw size | Estimated transfer size

chunk-S454IU5R.js | enrollment-routes | 11.87 kB | 2.99 kB

Application bundle generation complete. [12.205 seconds] - 2026-07-07T03:50:29.485Z

Output location: C:\Users\HP\Desktop\Deepskilling\Angular_HandsOn\student-course-portal\dist\student-course-portal

PS C:\Users\HP\Desktop\Deepskilling\Angular_HandsOn\student-course-portal>